ESE5023 Assignment 01 Report

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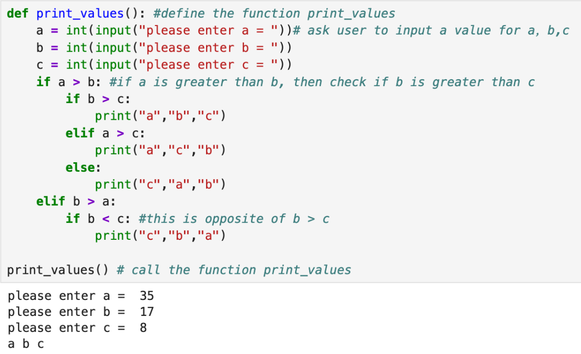
- ## PS1_1

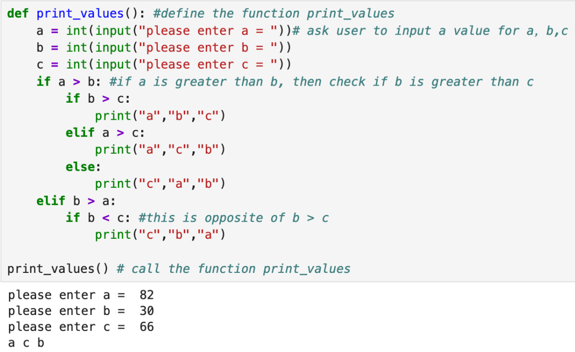
I enter several values of a, b, c to print different outputs, as shown in the following figures:



```python
def print_values(): #define the function print_values
    a = int(input("please enter a = "))# ask user to input a value for a, b,c
    b = int(input("please enter b = "))
    c = int(input("please enter c = "))
    if a > b: #if a is greater than b, then check if b is greater than c
        if b > c:
            print("a","b","c")
        elif a > c:
            print("a","c","b")
        else:
            print("c","a","b")
    elif b > a:
        if b < c: #this is opposite of b > c
            print("c","b","a")

print_values() # call the function print_values
```

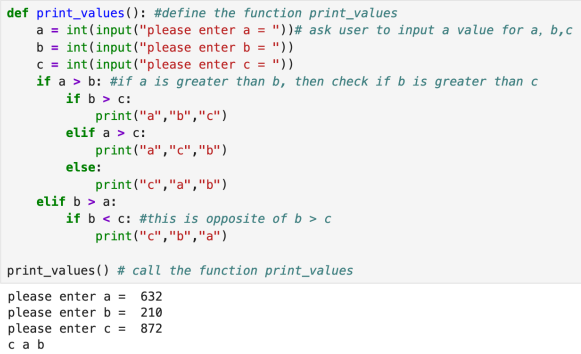
```
please enter a =  35
please enter b =  17
please enter c =  8
a b c
```



```python
def print_values(): #define the function print_values
    a = int(input("please enter a = "))# ask user to input a value for a, b,c
    b = int(input("please enter b = "))
    c = int(input("please enter c = "))
    if a > b: #if a is greater than b, then check if b is greater than c
        if b > c:
            print("a","b","c")
        elif a > c:
            print("a","c","b")
        else:
            print("c","a","b")
    elif b > a:
        if b < c: #this is opposite of b > c
            print("c","b","a")

print_values() # call the function print_values
```

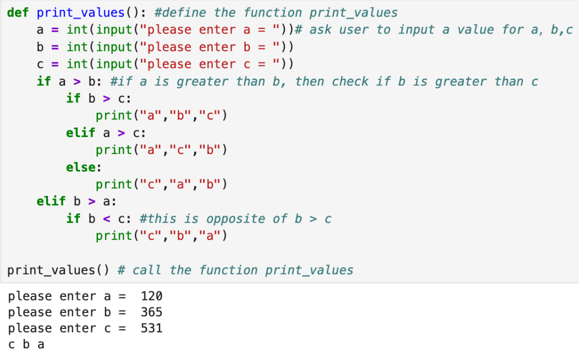
```
please enter a =  82
please enter b =  30
please enter c =  66
a c b
```



```python
def print_values(): #define the function print_values
    a = int(input("please enter a = "))# ask user to input a value for a, b,c
    b = int(input("please enter b = "))
    c = int(input("please enter c = "))
    if a > b: #if a is greater than b, then check if b is greater than c
        if b > c:
            print("a","b","c")
        elif a > c:
            print("a","c","b")
        else:
            print("c","a","b")
    elif b > a:
        if b < c: #this is opposite of b > c
            print("c","b","a")

print_values() # call the function print_values
```

```
please enter a =  632
please enter b =  210
please enter c =  872
c a b
```



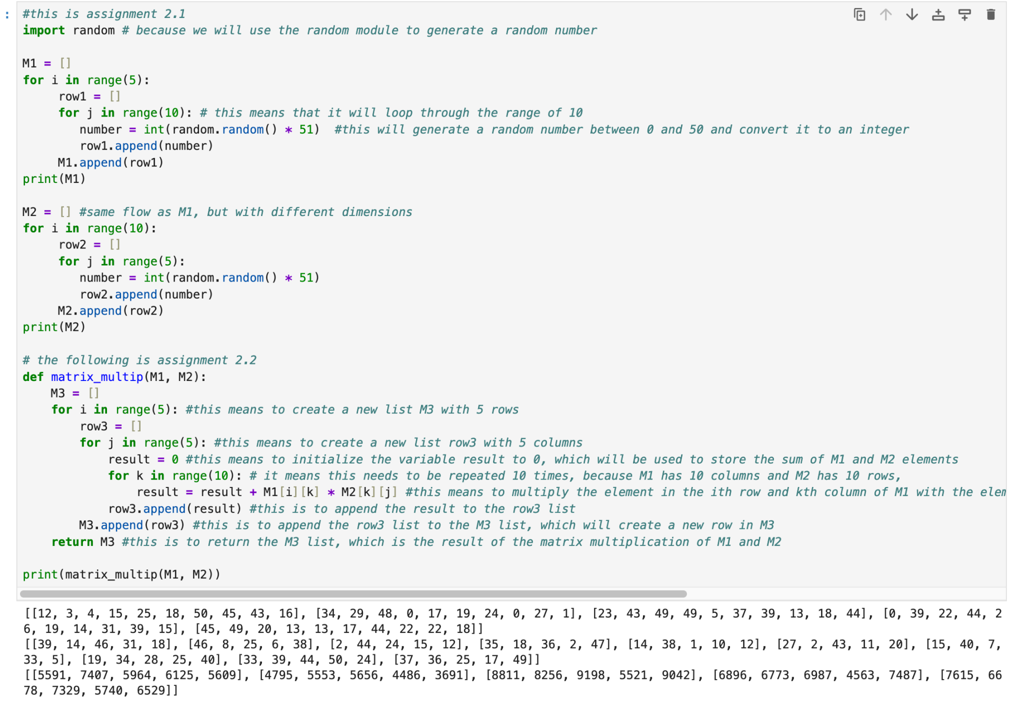
```python
def print_values(): #define the function print_values
    a = int(input("please enter a = "))# ask user to input a value for a, b,c
    b = int(input("please enter b = "))
    c = int(input("please enter c = "))
    if a > b: #if a is greater than b, then check if b is greater than c
        if b > c:
            print("a","b","c")
        elif a > c:
            print("a","c","b")
        else:
            print("c","a","b")
    elif b > a:
        if b < c: #this is opposite of b > c
            print("c","b","a")

print_values() # call the function print_values
```

```
please enter a =  120
please enter b =  365
please enter c =  531
c b a
```

Some notes are commented near the code.

- ## PS1_2



```python
#this is assignment 2.1
import random # because we will use the random module to generate a random number

M1 = []
for i in range(5):
     row1 = []
     for j in range(10): # this means that it will loop through the range of 10
        number = int(random.random() * 51)  #this will generate a random number between 0 and 50 and convert it to an integer
        row1.append(number)
     M1.append(row1)
print(M1)

M2 = [] #same flow as M1, but with different dimensions
for i in range(10):
     row2 = []
     for j in range(5):
        number = int(random.random() * 51)
        row2.append(number)
     M2.append(row2)
print(M2)

# the following is assignment 2.2
def matrix_multip(M1, M2):
    M3 = []
    for i in range(5): #this means to create a new list M3 with 5 rows
        row3 = []
        for j in range(5): #this means to create a new list row3 with 5 columns
            result = 0 #this means to initialize the variable result to 0, which will be used to store the sum of M1 and M2 elements
            for k in range(10): # it means this needs to be repeated 10 times, because M1 has 10 columns and M2 has 10 rows,
                result = result + M1[i][k] * M2[k][j] #this means to multiply the element in the ith row and kth column of M1 with the elem
            row3.append(result) #this is to append the result to the row3 list
        M3.append(row3) #this is to append the row3 list to the M3 list, which will create a new row in M3
    return M3 #this is to return the M3 list, which is the result of the matrix multiplication of M1 and M2

print(matrix_multip(M1, M2))
```

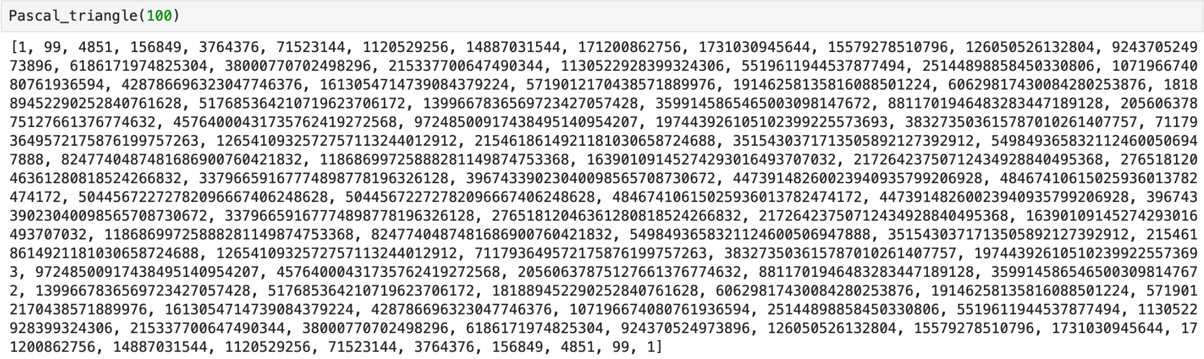
```
[[12, 3, 4, 15, 25, 18, 50, 45, 43, 16], [34, 29, 48, 0, 17, 19, 24, 0, 27, 1], [23, 43, 49, 49, 5, 37, 39, 13, 18, 44], [0, 39, 22, 44, 2
6, 19, 14, 31, 39, 15], [45, 49, 20, 13, 13, 17, 44, 22, 22, 18]]
[[39, 14, 46, 31, 18], [46, 8, 25, 6, 38], [2, 44, 24, 15, 12], [35, 18, 36, 2, 47], [14, 38, 1, 10, 12], [27, 2, 43, 11, 20], [15, 40, 7,
33, 5], [19, 34, 28, 25, 40], [33, 39, 44, 50, 24], [37, 36, 25, 17, 49]]
[[5591, 7407, 5964, 6125, 5609], [4795, 5553, 5656, 4486, 3691], [8811, 8256, 9198, 5521, 9042], [6896, 6773, 6987, 4563, 7487], [7615, 66
78, 7329, 5740, 6529]]
```

Here I reviewed the rule of matrix multiplication by online source:

https://mp.weixin.qq.com/s?__biz=MzI3MzE3OTI0Mw==&mid=2247581408&idx=1&sn=478f9b77e92750abb7794498aa228eac&chksm=ea1f4e85739f66a2e4be78925da1d0ebfc8486769e93827319057484f3d80447a1d2c867fb6a&scene=27

- **PS1_3**

This this Pascal triangle 100:



```
Pascal_triangle(100)
```

```
[1, 99, 4851, 156849, 3764376, 71523144, 1120529256, 14887031544, 171200862756, 1731030945644, 15579278510796, 126050526132804, 9243705249
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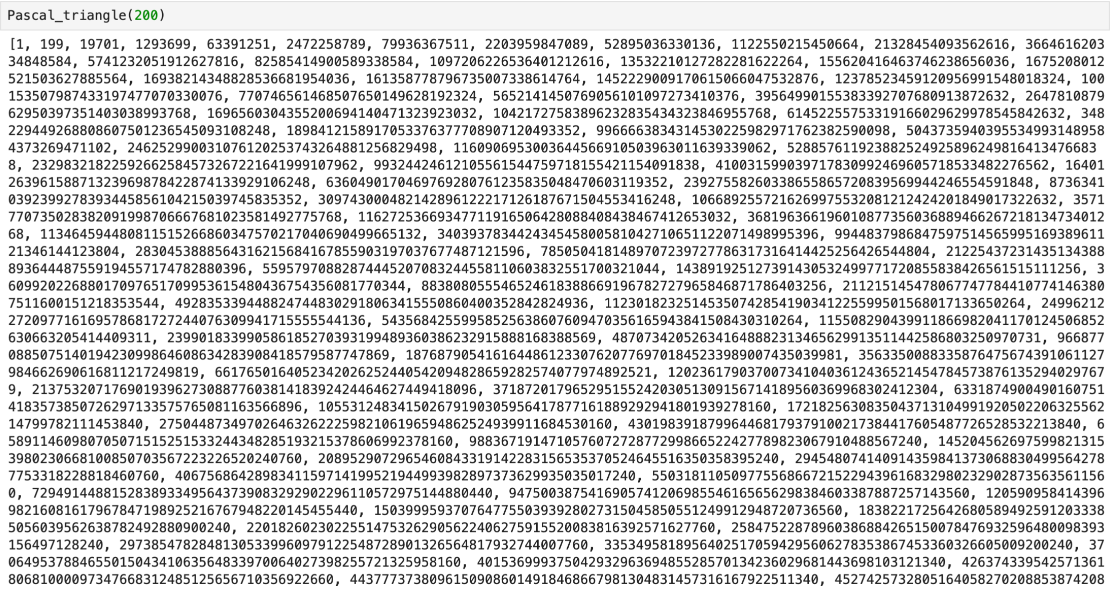
1200862756, 14887031544, 1120529256, 71523144, 3764376, 156849, 4851, 99, 1]
```

This this Pascal triangle 200:



```
Pascal_triangle(200)
```

```
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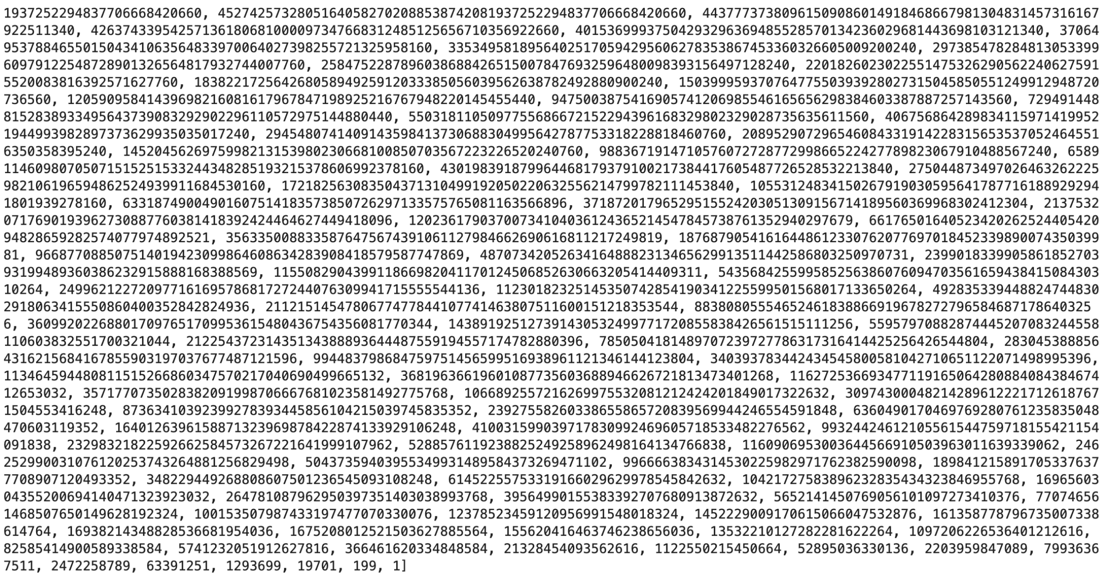

```
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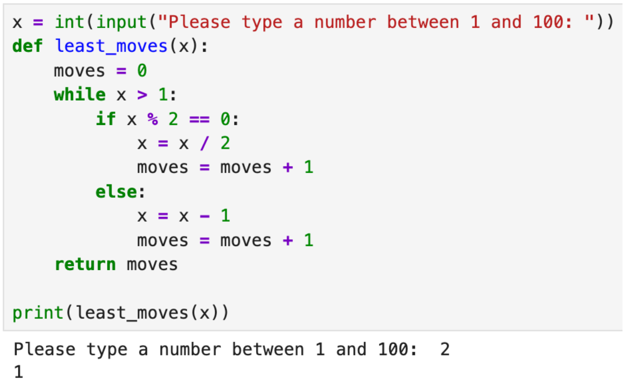
7511, 2472258789, 63391251, 1293699, 19701, 199, 1]
```

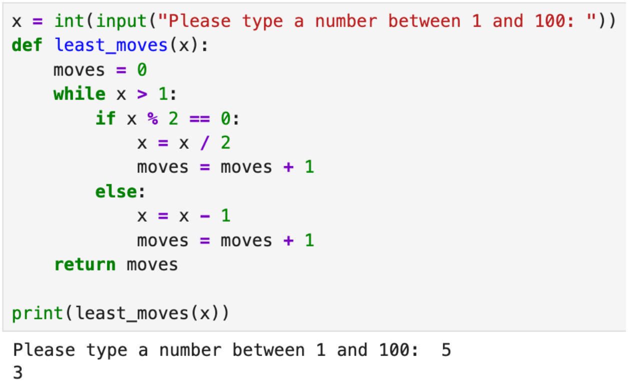
- ## PS1_4



```
x = int(input("Please type a number between 1 and 100: "))
def least_moves(x):
    moves = 0
    while x > 1:
        if x % 2 == 0:
            x = x / 2
            moves = moves + 1
        else:
            x = x - 1
            moves = moves + 1
    return moves

print(least_moves(x))
```

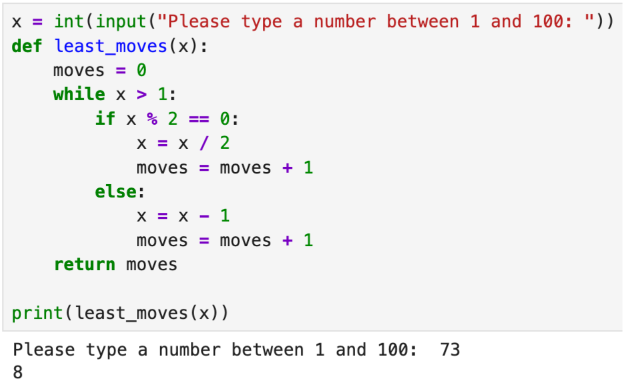
```
Please type a number between 1 and 100:  2
1
```



```
x = int(input("Please type a number between 1 and 100: "))
def least_moves(x):
    moves = 0
    while x > 1:
        if x % 2 == 0:
            x = x / 2
            moves = moves + 1
        else:
            x = x - 1
            moves = moves + 1
    return moves

print(least_moves(x))
```

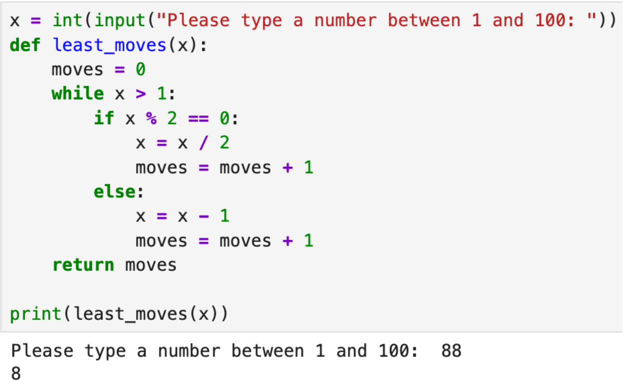
```
Please type a number between 1 and 100:  5
3
```



```
x = int(input("Please type a number between 1 and 100: "))
def least_moves(x):
    moves = 0
    while x > 1:
        if x % 2 == 0:
            x = x / 2
            moves = moves + 1
        else:
            x = x - 1
            moves = moves + 1
    return moves

print(least_moves(x))
```

```
Please type a number between 1 and 100:  73
8
```



```
x = int(input("Please type a number between 1 and 100: "))
def least_moves(x):
    moves = 0
    while x > 1:
        if x % 2 == 0:
            x = x / 2
            moves = moves + 1
        else:
            x = x - 1
            moves = moves + 1
    return moves

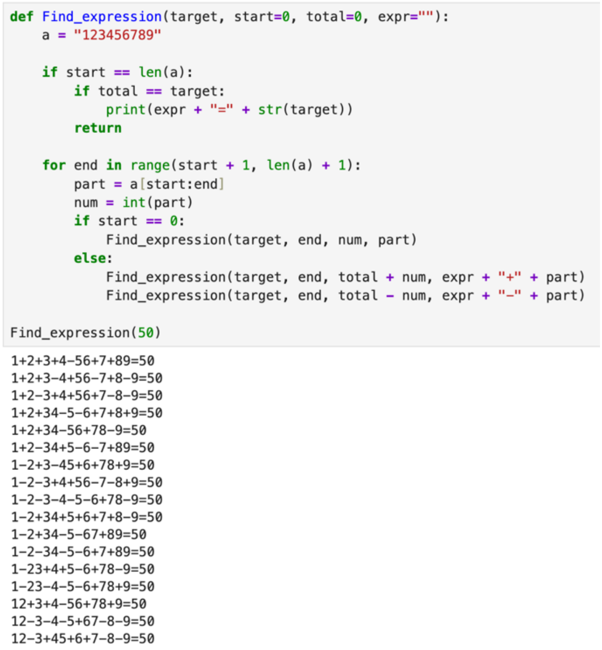
print(least_moves(x))
```

```
Please type a number between 1 and 100:  88
8
```

The first 2 outputs are to verify the result given in the assignment, detailed notes about the code could be found near the scripts.

- ## PS1_5

  I got explained about how to understand and break these questions in to parts by my friend Haochen, Yuan, who is also a PhD from PolyU.



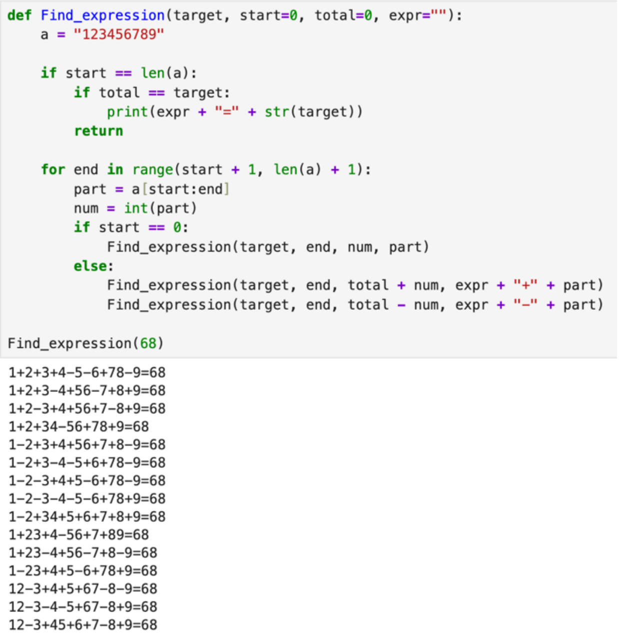
```
def Find_expression(target, start=0, total=0, expr=""):
    a = "123456789"

    if start == len(a):
        if total == target:
            print(expr + "=" + str(target))
        return

    for end in range(start + 1, len(a) + 1):
        part = a[start:end]
        num = int(part)
        if start == 0:
            Find_expression(target, end, num, part)
        else:
            Find_expression(target, end, total + num, expr + "+" + part)
            Find_expression(target, end, total - num, expr + "-" + part)

Find_expression(50)
```

```
1+2+3+4-56+7+89=50
1+2+3-4+56-7+8-9=50
1+2-3+4+56+7-8-9=50
1+2+34-5-6+7+8+9=50
1+2+34-56+78-9=50
1+2-34+5-6-7+89=50
1-2+3-45+6+78+9=50
1-2-3+4+56-7-8+9=50
1-2-3-4-5-6+78-9=50
1-2+34+5+6+7+8-9=50
1-2+34-5-67+89=50
1-2-34-5-6+7+89=50
1-23+4+5-6+78-9=50
1-23-4-5-6+78+9=50
12+3+4-56+78+9=50
12-3-4-5+67-8-9=50
12-3+45+6+7-8-9=50
```



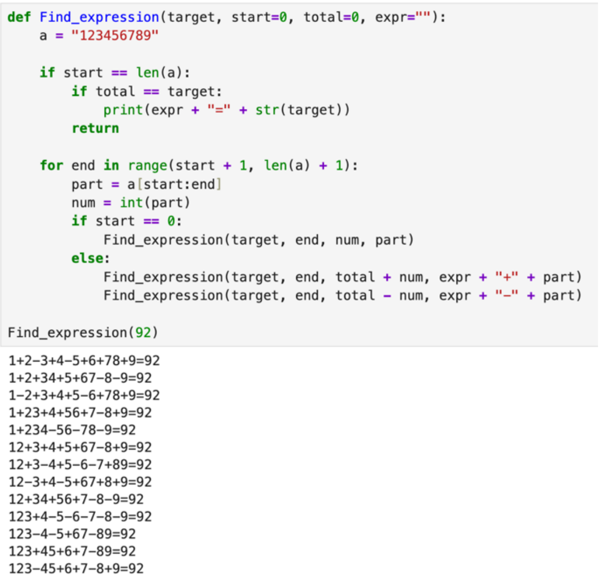
```
def Find_expression(target, start=0, total=0, expr=""):
    a = "123456789"

    if start == len(a):
        if total == target:
            print(expr + "=" + str(target))
        return

    for end in range(start + 1, len(a) + 1):
        part = a[start:end]
        num = int(part)
        if start == 0:
            Find_expression(target, end, num, part)
        else:
            Find_expression(target, end, total + num, expr + "+" + part)
            Find_expression(target, end, total - num, expr + "-" + part)

Find_expression(68)
```

```
1+2+3+4-5-6+78-9=68
1+2+3-4+56-7+8+9=68
1+2-3+4+56+7-8+9=68
1+2+34-56+78+9=68
1-2+3+4+56+7+8-9=68
1-2+3-4-5+6+78-9=68
1-2-3+4+5-6+78-9=68
1-2-3-4-5-6+78+9=68
1-2+34+5+6+7+8+9=68
1+23+4-56+7+89=68
1+23-4+56-7+8-9=68
1-23+4+5-6+78+9=68
12-3+4+5+67-8-9=68
12-3-4-5+67-8+9=68
12-3+45+6+7-8+9=68
```



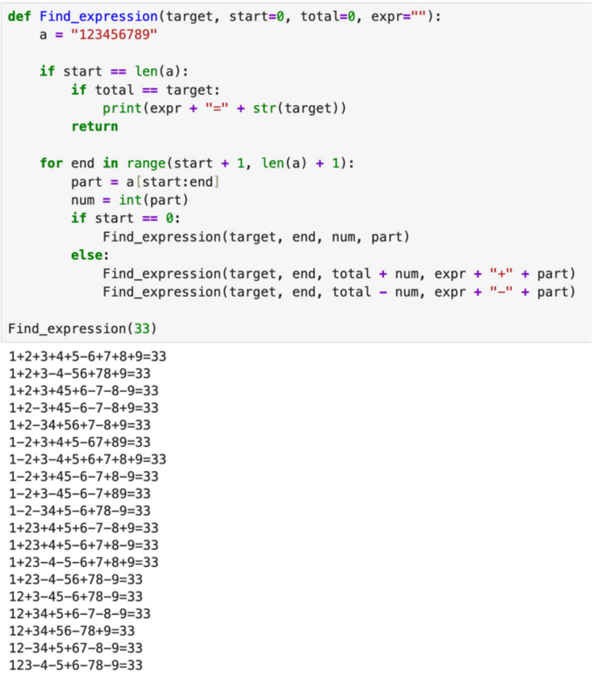
```python
def Find_expression(target, start=0, total=0, expr=""):
    a = "123456789"

    if start == len(a):
        if total == target:
            print(expr + "=" + str(target))
        return

    for end in range(start + 1, len(a) + 1):
        part = a[start:end]
        num = int(part)
        if start == 0:
            Find_expression(target, end, num, part)
        else:
            Find_expression(target, end, total + num, expr + "+" + part)
            Find_expression(target, end, total - num, expr + "-" + part)


Find_expression(92)
```

```
1+2-3+4-5+6+78+9=92
1+2+34+5+67-8-9=92
1-2+3+4+5-6+78+9=92
1+23+4+56+7-8+9=92
1+234-56-78-9=92
12+3+4+5+67-8+9=92
12+3-4+5-6-7+89=92
12-3+4-5+67+8+9=92
12+34+56+7-8-9=92
123+4-5-6-7-8-9=92
123-4-5+67-89=92
123+45+6+7-89=92
123-45+6+7-8+9=92
```



```python
def Find_expression(target, start=0, total=0, expr=""):
    a = "123456789"

    if start == len(a):
        if total == target:
            print(expr + "=" + str(target))
        return

    for end in range(start + 1, len(a) + 1):
        part = a[start:end]
        num = int(part)
        if start == 0:
            Find_expression(target, end, num, part)
        else:
            Find_expression(target, end, total + num, expr + "+" + part)
            Find_expression(target, end, total - num, expr + "-" + part)


Find_expression(33)
```

```
1+2+3+4+5-6+7+8+9=33
1+2+3-4-56+78+9=33
1+2+3+45+6-7-8-9=33
1+2-3+45-6-7-8+9=33
1+2-34+56+7-8+9=33
1-2+3+4+5-67+89=33
1-2+3-4+5+6+7+8+9=33
1-2+3+45-6-7+8-9=33
1-2+3-45-6-7+89=33
1-2-34+5-6+78-9=33
1+23+4+5+6-7-8+9=33
1+23+4+5-6+7+8-9=33
1+23-4-5-6+7+8+9=33
1+23-4-56+78-9=33
12+3-45-6+78-9=33
12+34+5+6-7-8-9=33
12+34+56-78+9=33
12-34+5+67-8-9=33
123-4-5+6-78-9=33
```

Above are the results generated by the script, I choose four random numbers from 1 to 100, the first “if” condition is to end the loop, the second for loop is to index the numbers in the string and the if condition is to call the function internally, start ==0 index the first number “1”.